

1)

```

> Fertilizer_A=c(55, 60, 50, 57)
> Fertilizer_B=c(62, 68, 63, 60)
> Fertilizer_C=c(53, 53, 52, 54)
> Fertilizer_D=c(68, 70, 120, 5)
> platns=data.frame(Fertilizer_A,Fertilizer_B,Fertilizer_C,Fertilizer_D)
> plants
  FertilizerA FertilizerB FertilizerC FertilizerD
1          55          62          53          68
2          60          68          53          70
3          50          63          52         120
4          57          60          54           5

> tablenew=stack(plants[,c("FertilizerA","FertilizerB","FertilizerC","FertilizerD")])
> colnames(tablenew)=c("heights","Fertilizer")
> tablenew
  heights Fertilizer
1      55 FertilizerA
2      60 FertilizerA
3      50 FertilizerA
4      57 FertilizerA
5      62 FertilizerB
6      68 FertilizerB
7      63 FertilizerB
8      60 FertilizerB
9      53 FertilizerC
10     53 FertilizerC
11     52 FertilizerC
12     54 FertilizerC
13     68 FertilizerD
14     70 FertilizerD
15    120 FertilizerD
16      5 FertilizerD

```

Normality checking

H0: Data are normally distributed

H1: Data are not normally distributed

```

> attach(tablenew)
> shapiro.test(heights)

```

Shapiro-wilk normality test

```

data: heights
W = 0.76704, p-value = 0.001027

```

Since $p\text{-value} < 0.05$, H_0 is rejected at 5% level of significance. Therefore the data are not normally distributed. So we have to perform suitable non-parametric test.

Hypothesis

H0: The distributions of plant heights are identical across all fertilizer groups. (There is no significant difference in the mean plant heights among the four fertilizer treatments.)

H1: At least one fertilizer treatment produces systematically different plant heights than the others. (There is a significant difference in the mean plant heights among the four fertilizer treatments.)

```
> kruskal.test(heights~Fertilizer,data=tablenew)
```

Kruskal-wallis rank sum test

data: heights by Fertilizer

Kruskal-wallis chi-squared = 6.5196, df = 3, p-value = 0.08889

Since p value > 0.05, H0 do not reject at 5% level of significance. Therefore there is no evidence that the fertilizer has a significant effect on plant height at the 5% level of significance.

2)

```
> MethodA=c(72, 75, 78, 74, 76)
> MethodB=c(68, 70, 72, 69, 71)
> MethodC=c(80, 82, 85, 83, 81)
> MethodD=c(65, 67, 66, 64, 68)
> riceyield=data.frame(MethodA,MethodB,MethodC,MethodD)
> riceyield
```

	MethodA	MethodB	MethodC	MethodD
1	72	68	80	65
2	75	70	82	67
3	78	72	85	66
4	74	69	83	64
5	76	71	81	68

```
> irrigation=stack(riceyield[,c("MethodA","MethodB","MethodC","MethodD")])
> colnames(irrigation)=c("kg_perplot","irrigation_method")
> irrigation
```

	kg_perplot	irrigation_method
1	72	MethodA
2	75	MethodA
3	78	MethodA
4	74	MethodA
5	76	MethodA
6	68	MethodB
7	70	MethodB
8	72	MethodB
9	69	MethodB
10	71	MethodB
11	80	MethodC
12	82	MethodC
13	85	MethodC

222116

14	83	MethodC
15	81	MethodC
16	65	MethodD
17	67	MethodD
18	66	MethodD
19	64	MethodD
20	68	MethodD

a)

Normality checking

H0: Data are normally distributed

H1: Data are not normally distributed

```
> shapiro.test(kg_perplot)
```

Shapiro-wilk normality test

data: kg_perplot

W = 0.94742, p-value = 0.3296

Since p value > 0.05, H0 is not rejected at 5% level of significance. Therefore the data are normally distributed. So we have to perform ANOVA test.

Hypothesis

H0: irrigation method has no significant effect on rice crop yield.

H1: irrigation method has a significant effect on rice crop yield.

```
> attach(irrigation)
```

```
> model1=aov(kg_perplot~irrigation_method,data=irrigation)
```

```
> summary(model1)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
irrigation_method	3	731.4	243.80	71.18	1.8e-09 ***
Residuals	16	54.8	3.43		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Since P value < 0.05, H0 is rejected at 5% level of significance. Therefore we can conclude that irrigation method has an effect on crop yield at the 5% level of significance.

b)

Hypothesis

$$H_0: \mu_i = \mu_j$$

$$H_1: \mu_i \neq \mu_j \quad ; \text{ where } \mu_i = \text{mean rice kg per plot when using different irrigation techniques and } i \& j \in \{A, B, C, D\}$$

```
> TukeyHSD(model1)
```

```
Tukey multiple comparisons of means
95% family-wise confidence level
```

```
Fit: aov(formula = kg_perplot ~ irrigation_method, data = irrigation)
```

```
$irrigation_method
      diff      lwr      upr      p adj
MethodB-MethodA -5.0 -8.348738 -1.6512622 0.0029468
MethodC-MethodA  7.2  3.851262 10.5487378 0.0000748
MethodD-MethodA -9.0 -12.348738 -5.6512622 0.0000051
MethodC-MethodB 12.2  8.851262 15.5487378 0.0000001
MethodD-MethodB -4.0 -7.348738 -0.6512622 0.0167017
MethodD-MethodC -16.2 -19.548738 -12.8512622 0.0000000
```

Since the all the p values less than 0.05, we can reject H_0 at 5% level of significance. Therefore we can conclude that mean rice kg per plot when using each irrigation techniques are significantly different.

c)

Normality checking

H_0 : residuals are normally distributed.

H_1 : residuals are not normally distributed.

```
> gg=residuals(model1)
> shapiro.test(gg)
```

```
Shapiro-wilk normality test
```

```
data: gg
```

```
W = 0.96784, p-value = 0.7087
```

Since P value > 0.05, we do not reject H_0 at 5% level of significance. Therefore residuals are normally distributed .

Homogeneity checking

H_0 : residuals have constant variance

H_1 : residuals haven't constant variance

```
> leveneTest(model1)
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  3   0.163 0.9197
      16
```

Since P value > 0.05, we do not reject H₀ at 5% level of significance. Therefore residuals have constant variance.

3)

```
> drug=read.table(file.choose(),header=T)
> newdrug=stack(drug[,c("Drug_A","Drug_B","Drug_C","Drug_D")])
> colnames(newdrug)=c("blood_pressure","drug_type")
```

```
> newdrug
      blood_pressure drug_type
1             12     Drug_A
2             15     Drug_A
3             14     Drug_A
4             13     Drug_A
5             16     Drug_A
6             14     Drug_A
7             18     Drug_B
8             20     Drug_B
9             19     Drug_B
10            17     Drug_B
11            21     Drug_B
12            18     Drug_B
13            10     Drug_C
14            11     Drug_C
15             9     Drug_C
16            12     Drug_C
17            10     Drug_C
18            11     Drug_C
19            14     Drug_D
20            16     Drug_D
21            15     Drug_D
22            13     Drug_D
23            17     Drug_D
24            16     Drug_D
```

A.

Normality checking

H₀: Data are normally distributed

H₁: Data are not normally distributed

```
> shapiro.test(blood_pressure)
```

Shapiro-wilk normality test

```
data: blood_pressure
w = 0.97702, p-value = 0.8352
```

Since $p\text{-value} > 0.05$, H_0 is not rejected at 5% level of significance. Therefore the data are normally distributed. So we have to perform ANOVA test.

Hypothesis

H_0 : There is no significant difference in effectiveness among the drugs.

H_1 : There is a significant difference in effectiveness among the drugs.

```
> model2=aov(blood_pressure~drug_type,data=newdrug)
> summary(model2)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
drug_type	3	212.46	70.82	38.11	1.84e-08 ***
Residuals	20	37.17	1.86		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Since $P\text{-value} < 0.05$, H_0 is rejected at 5% level of significance. Therefore we can conclude that there is a significant difference in effectiveness among the drugs.

B. Hypothesis

$H_0: \mu_i = \mu_j$

$H_1: \mu_i \neq \mu_j$;

Where μ_i = mean blood pressure when using different drug type and $i \& j \in \{A, B, C, D\}$

```
> TukeyHSD(model2)
Tukey multiple comparisons of means
95% family-wise confidence level
```

```
Fit: aov(formula = blood_pressure ~ drug_type, data = newdrug)
```

```
$drug_type
```

	diff	lwr	upr	p adj
Drug_B-Drug_A	4.833333	2.630436	7.036230	0.0000295
Drug_C-Drug_A	-3.500000	-5.702897	-1.297103	0.0013050
Drug_D-Drug_A	1.166667	-1.036230	3.369564	0.4660167
Drug_C-Drug_B	-8.333333	-10.536230	-6.130436	0.0000000
Drug_D-Drug_B	-3.666667	-5.869564	-1.463770	0.0008050

222116

Drug_D-Drug_C 4.666667 2.463770 6.869564 0.0000468

From the output ,

Since only the P value of Drug_D-Drug_A is greater than 0.05 , therefore H0 is not rejected for that pair.

Therefore the mean blood pressure using drug A equal to using drug D.

Since the all other the p values less than 0.05, we can reject H0 at 5% level of significance for those pair

Therefore we can conclude that mean blood pressure when using each drug type are significantly different.

C.

Normality checking

H0: residuals are normally distributed.

H1: residuals are not normally distributed.

```
> yy=residuals(model2)
> shapiro.test(yy)
```

Shapiro-wilk normality test

data: yy

w = 0.97109, p-value = 0.6939

Since P value>0.05, we do not reject H0 at 5% level of significance. Therefore residuals are normally distributed .

Homogeneity checking

H0: residuals have constant variance

H1: residuals haven't constant variance

```
> leveneTest(model2)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	3	0.2546	0.8571
	20		

Since P value>0.05, we do not reject H0 at 5% level of significance. Therefore residuals have constant variance.

4)

```

> treatment_1=c( 55, 60, 58, 57);
> treatment_2=c( 62, 65, 63, 66);
> treatment_3=c( 51, 53, 52, 54);
> treatment_4=c( 68, 95, 40, 120);
> Treat=data.frame(treatment_1,treatment_2,treatment_3,treatment_4)

> Treatment=stack(Treat[,c("treatment_1","treatment_2","treatment_3","treatment_4")])
> colnames(Treatment)=c("data","treatment_type")
> Treatment
  data treatment_type
1    55    treatment_1
2    60    treatment_1
3    58    treatment_1
4    57    treatment_1
5    62    treatment_2
6    65    treatment_2
7    63    treatment_2
8    66    treatment_2
9    51    treatment_3
10   53    treatment_3
11   52    treatment_3
12   54    treatment_3
13   68    treatment_4
14   95    treatment_4
15   40    treatment_4
16  120    treatment_4

```

I.

Hypothesis

H0: There is no significant difference in the mean responses among the four treatment groups

H1: There is a significant difference in the mean responses among the four treatment groups

```

> model3=aov(data~treatment_type,data=Treatment)
> summary(model3)

```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treatment_type	3	1819	606.2	2.024	0.164
Residuals	12	3595	299.6		

Since p value > 0.05 , therefore H_0 is not rejected at 5% level of significance. Therefore we can conclude that There is no significant difference in the mean responses among the four treatment groups.

II.

Normality checking

H_0 : residuals are normally distributed.

H_1 : residuals are not normally distributed.

```
> xx=residuals(model3)
> shapiro.test(xx)
```

Shapiro-wilk normality test

```
data: xx
W = 0.75481, p-value = 0.0007263
```

Since P value < 0.05 , we reject H_0 at 5% level of significance. Therefore residuals are not normally distributed .

Homogeneity checking

H_0 : residuals have constant variance

H_1 : residuals haven't constant variance

```
> leveneTest(model3)
Levene's Test for Homogeneity of Variance (center = median)
  Df F value    Pr(>F)
group 3   10.88 0.0009706 ***
  12
```

Since P value < 0.05 , we reject H_0 at 5% level of significance. Therefore residuals have not constant variance.

III.

An alternative method for violating homogeneity

```
> oneway.test(data~ treatment_type, data =Treatment)
```

One-way analysis of means (not assuming equal variances)

```
data: data and treatment_type
F = 29.896, num df = 3.0000, denom df = 6.2683, p-value =
0.0004196
```

Again, Since P value < 0.05 , we reject H_0 at 5% level of significance. Therefore residuals have not constant variance.